



THE COPPERBELT UNIVERSITY

MICHAEL CHILUFYA SATA SCHOOL OF MEDICINE

END OF TERM 1 TEST - MARCH, 2021

MBS230 - MEDICAL BIOCHEMISTRY-I

Student ID (Computer No.) : _____
Name : _____
Program of Study : _____
Duration of Examination : 02 hours
Total Marks : 60 marks

INSTRUCTIONS TO CANDIDATES

READ CAREFULLY BEFORE YOU START TO ANSWER

- Write your STUDENT IDENTIFICATION NUMBER (SIN) and NAMES on the spaces provided on the front page.
- Read the instructions at the beginning of every section (Sections A and B). Answer all questions. No negative marking applies for wrong answers in Section A. Mark allocations for questions in Section B are shown in brackets.
- USE ONLY BLUE OR BLACK INK PEN to answer. Other colours may be used for diagrams only. Answers written in pencil shall not be considered, except for diagrams.
- Answer in this paper only. For section A, circle the letter corresponding to the correct answer and for section B, your answer should be in the provided space. Your answers should be LEGIBLE AND READABLE, otherwise they shall not be considered.
- No rough work is allowed anywhere in the paper, except on the last page i.e specifically allocated space. If any kind of rough work is found on any page other than on the space specifically allocated, the page (s) shall not be considered.
- No one is shall leave the examination room in the first 30 minutes and in the last 30 minutes of examination.

SECTION - A: MULTIPLE CHOICE QUESTIONS - CHOOSE ONLY ONE CORRECT AND BEST ANSWER FOR EACH QUESTION AND CIRCLE THE LETTER IN FRONT OF IT. ANSWER ALL QUESTIONS (40x.5 = 20 MARKS)

1. Which one of the following statements concerning glutamine is correct?

- ☒ a. Contains an amide group
- b. Has Glu as its three letter abbreviation
- c. Classified as an acidic amino acid
- d. Has a charged polar side chain
- e. Doesn't have a chiral carbon

2. Which amino acid is capable of forming disulfide bonds?

- a. Glycine
- b. Proline
- c. Histidine
- ☒ d. Cysteine
- e. Isoleucine

3. amino acid has 2 chiral carbons

- ☒ a. Isoleucine
- b. Valine
- c. Proline
- d. Leucine
- e. Glycine

4. is associated with a condition known as phenylketonuria.

- a. Proline
- b. Methionine
- ☒ c. Phenylalanine
- d. Histidine
- e. Valine

5. An amino acid that yields acetoacetyl CoA during catabolism of its carbon skeleton is.....

- a. Glucogenic
- ☒ b. Ketogenic
- c. Hydrophobic
- d. Polar
- e. Polar and hydrophobic

6. is an amino acid the contains an imidazole aromatic ring

- a. Tyrosine
- b. Phenylalanine
- ☒ c. Histidine
- d. Proline
- e. Tryptophan

7. exist in two non superimposable mirror images of each other
- a. Anomers
 - b. Epimers
 - ☒ c. Enantiomers
 - d. Mutarotors
 - e. Chiral carbons
8. mainly contribute to stabilisation of secondary structures of the proteins.
- a. Hydrophobic bonds
 - b. Van der waal forces
 - c. Salt bridges
 - ☒ d. Hydrogen bonds
 - e. Peptide bonds
9. A chiral carbon of an amino acid arises from the fact that it.....
- a. is symmetric
 - b. is bonded to both amino and carboxylic groups
 - ☒ c. is bonded to four different chemical groups
 - d. has net charge of zero
 - e. assumes an L configuration
10. is an element found in all amino acids but it is not found in all carbohydrates.
- a. Oxygen
 - b. Carbon
 - c. Hydrogen
 - ☒ d. Nitrogen
 - e. Sulphur
11. A peptide bond is formed between.....
- a. two carboxyl groups
 - b. two aromatic groups
 - ☒ c. a carboxyl group and an amino group
 - d. an ester and an amine group
 - e. two amino groups
12. Most of the amino acids found in human body are.....
- ☒ a. L-isomers
 - b. D-isomers
 - c. Optical isomers
 - d. D and L- isomers
 - e. optically inactive
13. primarily determines the function of a protein
- a. Number of amino acids
 - ☒ b. Spatial conformation
 - c. Molecular weight
 - d. Chirality of the carbon atoms
 - e. Affinity for water

14. Which one of the following amino acids is found in the β - bends as kinks?

- a. Glycine
- b. Alanine
- c. Glutamic acid
- ☒ d. Proline
- e. Valine

15. Functional groups of amino acids are called.....

- a. side chains
- ☒ b. amino and carboxylic acid
- c. peptides
- d. polymers
- e. phosphorylated side chains

16. All of the following statements are correct regarding peptides except.....

- a. It contains amino terminus
- b. It contains carboxyl terminus
- c. It contains peptide bonds
- ☒ d. It contains only basic amino acids
- e. It contains between 2 and 50 amino acids

17. Which of the following is not correct?

- ☒ a. Protoporphyrin is bound to a single iron atom in its Fe^{+3} state
- b. Iron atom has 6 coordination bonds
- c. Heme consists of a complex organic ring structure, protoporphyrin
- d. Heme is found in a number of oxygen transporting proteins
- e. One of the positions is coordinated to the side chain of a histidine residue

18. Myoglobin is mostly abundant in the.....

- a. Skin
- b. Bones
- ☒ c. Muscles
- d. Nerves
- e. White blood cells

19. Which of these statements is correct about the role of heme in cytochromes?

- a. Functions as part of the active site
- b. Binds oxygen irreversibly
- ☒ c. Functions as an electron carrier
- d. Functions as a competitive inhibitor in reactions
- e. Functions as an activator

20. Heme is a complex of.....

- a. protoporphyrin IX and ferric iron
- ☒ b. protoporphyrin IX and ferrous iron
- c. porphyrin IX and ferric iron
- d. protoporphyrin IX and Histidine
- e. porphyrin IX and ferrous iron

21. is due to the oxidation of heme component to the ferric state

- a. Myoglobin
- ☒ b. Metmyoglobin
- c. Hemoglobin C
- d. Hemoglobin F
- e. Hemoglobin SC

22. *NADH-Cytochrome b₅ reductase*, an enzyme present in red blood cells, helps to correct the occasional oxidation of the hemoglobin iron. Why are newborns particularly susceptible to methemoglobin-producing compounds?

- a. New borns lack the enzyme
- ☒ b. The capacity of the enzyme is lower
- c. Their hemoglobin lacks the α -globin chain
- d. They only have γ -globin chains
- e. The enzyme at this stage has various mutations

23. Phenylketonuria is a genetic disorder that results from.....

- ☒ a. low or absent activity of an enzyme phenylalanine hydroxylase
- b. low or absent activity of an enzyme tyrosinase
- c. low or absent activity of an enzyme tyrosine oxidase
- d. low or absent activity of an enzyme phenyl oxidase
- e. low or absent activity of an enzyme that converts a non essential amino acid to an essential amino acid

24. Essential amino acids are.....

- a. synthesized by the body
- ☒ b. obtained from the diet
- c. not involved in the formation of peptides
- d. optically inactive
- e. all acidic

25. The carbon atom to which groups in an amino acid are bonded is called alpha (α) because it is.....

- ☒ a. adjacent to the carboxyl (acidic) group
- b. adjacent to the amino (basic) group
- c. optically active
- d. adjacent to a hydrogen atom
- e. optically inactive

26. The portion of a transmembrane protein that is located within the plasma membrane is most likely to be composed of amino acids that are.....

- a. acidic
- b. basic
- ☒ c. hydrophobic
- d. glycosylated
- e. acetylated

Which one of the following is referred to as an allosteric effector in the binding of oxygen to hemoglobin reversibly?

- ☒ a. pH of the environment
- b. presence of nitrogen
- c. presence of water
- d. presence of myoglobin
- e. presence of Lead

28.refers to glycosylated Hemoglobin

- a. Hb Ac
- b. HbA₂
- c. HbA_b
- d. Hb1Ac
- ☒ e. HbA_{1c}

29. Name the globin chains that are affected in glycosylated Hemoglobin

- a. γ -globin chains
- b. α -globin chains
- ☒ c. β -globin chains
- d. δ -globin chains
- e. both γ and α -globin chains

30. Patients homozygous for hemoglobin C have a condition that arises from

- ☒ a. substitution of lysine by glutamic acid in the β -chain
- b. substitution of glutamic acid by lysine in the β -chain
- c. substitution of valine by glutamic acid in the β -chain
- d. substitution of a polar amino acid by a non-polar one in the β -chain
- e. substitution of one of the α -chains by a β -chain

31. Which one of these statements is correct about thalassemias?

- ☒ a. hereditary hemolytic diseases in which there is an imbalance in the synthesis of globin chains
- b. they are polygenic disorders
- c. only α -globin chains are affected
- d. they are not in a class of hemoglobinopathies
- e. Only γ -chains are affected

32. Which two amino acids in collagen usually undergo post translational modification?

- a. lysine and histidine
- b. glycine and alanine
- c. valine and isoleucine
- ☒ d. lysine and proline
- e. leucine and glutamic acid

33. Lack of post translational modifications of some amino acids in collagen leads to non-cross linkage of the fibers. Which one of the following refers to the condition that arises?

- a. sickle cell anemia
- b. pernicious anemia
- ☒ c. scurvy
- d. phenylketonuria
- e. hemosiderosis

34. Which one of these statements is correct about the Ehlers-Danlos Syndrome?

- a. results from defective heme synthesis
- b. results from defective porphyrin ring synthesis
- ☒ c. results from defective collagen synthesis
- d. results from defective elastin synthesis
- e. results from defective keratin synthesis

35. is the type of fibrous protein found in hair, nails and outer epidermal layer of mammals

- a. Elastin
- ☒ b. Keratin
- c. Collagen
- d. Procollagen
- e. Proelastin

36. is not an example of a heme protein

- a. Cytochrome P450
- b. Catalase
- c. Myoglobin
- d. Tryptophan pyrrolase
- ☒ e. δ -aminolevulinic acid (ALA)

37. Which non essential amino acid is involved in the formation of all the carbon and nitrogen atoms of the porphyrin ring?

- a. Phenyl alanine
- b. Histidine
- c. Tyrosine
- ☒ d. Glycine
- e. Alanine

38. The catabolism of hemoglobin.....

- a. results in the formation of protoporphyrinogen
- b. results in the formation carbon dioxide
- c. occurs in red blood cells
- d. is the only source of bilirubin
- ☒ e. involves the destruction of the porphyrin ring

39. Which of the following statements is correct about most enzymes?

- a. increase the activation energy
- ☒ b. can make a reaction happen faster
- c. are used once and need to be replaced after catalyzing a reaction
- d. they change the position of the reaction's equilibrium
- e. are not specific to the type of substrate they receive

10. Which of the following statements about sickle cell anemia is correct? It is.....

- a. an autosomal recessive disorder
- b. an autosomal dominant disorder
- c. a sex linked disorder
- d. due to increased solubility of Hemoglobin
- e. due to reduced deformability of red blood cells

12.5
21.5
340

SECTION - B: DESCRIPTIVE ANSWER TYPE/SCENARIO BASED QUESTIONS
ANSWER ALL QUESTIONS. FOR EACH ANSWER, ILLUSTRATIONS WITH
DIAGRAMS (WHERE POSSIBLE) WILL ATTRACT EXTRA MARKS (40 MARKS)

1. Describe the following models of enzyme action (4 marks)

i. Induced fit (2 marks)

① The shape of the active site of the enzyme is not complementary to that of the substrate. It becomes complementary only when substrate binds. Shape of active site is not fixed.

ii. Lock and Key (2 marks)

① Shape of the active site of the enzyme is complementary to that of the active site even before the substrate binds. Shape of active site is fixed.

2. Explain how a metabolic pathway can be self regulating i.e how it can keep itself off and on (1 mark)

When the end product of the pathway is in excess it binds to an allosteric site on enzyme 1 and changes the active site, rendering it incapable of binding the substrate.

3. Explain how denaturation of an enzyme affects its activity. State the effect of the denaturant on each of the structures of the protein enzyme. Do you envisage return of enzyme activity when the denaturant is removed? (2 marks)

Changes the conformation loss of secondary tertiary structures. No effect on primary structure
Yes, depending on the extent of denaturation

4. State what happens to the rate of product formation in an enzyme catalysed reaction if you continue to

i. add the substrate (1 mark)

increase until enzyme is saturated

ii add an inhibitor (1 mark)

decrease in product formation

iii an enzyme (1 mark)

increase for as long as the substrate is present

5. Describe the following, in reference to proteins (6 marks)

a. Primary structure (1 mark)

Sequence of amino acids joined by peptide bonds

b. Secondary structure (1 mark)

The α - and β -bonds β -pleated sheet structure held together by hydrogen bonds between peptide carbonyl oxygen and amide hydrogen

c. Tertiary structure (1 mark)

The folding of domains (basic unit of structure and function) and final arrangement of domains in the polypeptide tertiary structure

d. Quaternary structure (1 mark)

Arrangement of polypeptide subunits that are either identical or totally unrelated. Disulfide, hydrophobic, hydrogen bonds and ionic interactions stabilise these structures

e. What are chaperones and what is their role in protein structure? (2 marks)

① Proteins that assist in conformational folding or unfolding and assembly of protein structure

6. State two properties of optically active isomers (2 marks)

i. Rotate plane polarized light to the left or right

ii. Non-super imposable mirror images of each other

7. Explain the statement that an enzyme does not change the position of the equilibrium of the reaction (1 mark)

An enzyme does not make an unfavourable reaction favourable

8. What is the role of Methemoglobin in cyanide poisoning (2 marks)

Methemoglobin easily binds ^(high affinity) cyanide to form cyanomethemoglobin and so this reduces the poisonous effect of cyanide

9. Explain why increased amounts of glycosylated Hemoglobin is found in patients with Diabetes mellitus (2 marks)

About 5% of HbA is glycosylated in normal human beings. In diabetes, since the glucose level is high, it therefore follows that even more HbA will be glycosylated.

10. Explain how, in Hemoglobin, the substitution of the non polar amino acid for a charged amino acid residue in patients with sickle cell disease results in interruption of oxygen to tissues (2 marks)

The red blood cells tend to

11. Explain the selective advantage for sickle cell heterozygotes in as far as infection with *Plasmodium falciparum* is concerned (2 marks)

The parasite does not complete its life cycle inside the red blood cells as the red blood cells have a shorter half life.

12. Explain how the Michaelis-Menten's constant (K_m) relates to the affinity of the enzyme (2 marks)

The higher the K_m , the lower the affinity
The lower the K_m , the higher the affinity

① 13. Explain how cells are able to produce proteins with so many properties and activities from just 20 amino acids (2 marks)

This is possible because of so many different combinations or order of amino acids that are possible

① 14. Explain the classification of thalassemias as quantitative hemoglobinopathies (2 marks)

Because there is less of one or more globin chains

15. Enzymes are often used in clinical diagnosis, however one of the disadvantages to this, is lack of specificity for a particular tissue. Explain this observation in reference to an anti-TB drug, rifampicin (2 marks)

In patients taking rifampicin, an enzyme, GGT is elevated

16. One of the specialised products formed from amino acids, known as a Happy chemical, is Serotonin. It is found in several sites in the body. It is involved in normal and abnormal behaviour and sleep regulation. State the precursor amino acid and the type of reaction(s) that are involved in its formation (2 marks)

Tryptophan

- Hydroxylation and decarboxylation

17. Histamines are chemical messengers with a wider range of cellular response, among them, allergic, inflammatory. Which amino acid is a precursor for histamines and what type of a reaction takes place to produce histamines (2 marks)

Histidine

decarboxylation

18. Collagen is the most abundant fibrous protein. Describe its structure, stating the number of polypeptide chains and what variation, in the amino acid sequence, achieves it (2 marks)

3 polypeptides in a triple helix held together by hydrogen bonds - Elongated triple helical structure
aa side chains on the outside
Variations in amino acid sequences as aa have different properties

19. Explain an irreversible type of enzyme inhibition. Give examples and state the type of bonds the inhibitor will form with the enzyme (2 marks)

The inhibitor binds to the enzyme covalently.
Organophosphorous compounds - Nerve gases
cause permanent, chemical modification of
essential functional groups in the enzyme.

What is an allosteric activator in enzyme catalysis? (1 mark)

Binds on a site different from active site and facilitates the formation of a correct conformation of active site

i. How does an allosteric activator affect the active site? (1 mark)

It ensures the active site attains the correct conformation for the substrate to bind.

END OF THE TEST